

Graduation Project Proposal Form

1. Project Information

- **Project Title:** Gas Gap Solutions
- **GitHub Link:** <https://github.com/DEPI-Ismailia-G1-2025>
- **Course/Track:** Data Analysis - NXT2_ONL3_DAT1_G3
- **Team Members: Ismailia Group 1 – Total members (7)**
 1. Samy Abdel Salam Mahmoud (Team Leader)
 2. Hala Nour El-Deen Ali
 3. Gehad Khaled Abaas
 4. Ola Abdel Twab Sayed Ahmed
 5. Mahmoud Sami Abdullah
 6. Mahmoud Ebraik Abdou
 7. Mohamed Nasser Mohamed

2. Project Overview

- **Objective:** Provide an effective solution to the challenges Egyptians encounter when attempting to connect natural gas to their homes.
- **Scope of Work:** The scope of work is focused on Ismailia Governorate and its affiliated villages to Identify and analyze the key issues faced in the process of obtaining a natural gas connection for their homes.
- **Expected Outcomes:** Deliver data-driven recommendations aimed at eliminating or minimizing these issues to improve the gas connection process on real data analysis to save time, effort and costs for service beneficiaries.

3. Problem Statement

- The difficulty of the procedures that the Egyptians will have to go through when they are trying to have gas at their homes specially in Ismailia Governorate.

4. Proposed Solution

- **Technologies Used:**
 1. Data Collection & Cleaning: (Microsoft Excel – Power Query – Power Pivot – SQL) for structured data storage
 2. Data Processing & Analysis: (Microsoft Excel– Python) main language for data analysis.
 3. Visualization & Reporting: Tableau / Power BI – for dashboards
- **System Architecture:**
 1. Data Segmentations: Microsoft Excel - Power Query.
 2. Data Sources: Web scraping, APIs, CSV files
 3. Data Storage: SQL databases
 4. Data Processing: Cleaning, transformation
 5. Analysis: Modeling, clustering
 6. Visualization: Dashboards, charts, reports

5. Resources Needed

- **Personal:**

1. Project team members and recommendations from the team Instructor.
2. Data Controller in Natural Gas Company.

- **Hardware/Software:**

1. PCs, laptops, Printer and Scanner.
2. Microsoft Excel, SQL, Anaconda, Tableau and Power BI, Websites and collaboration software.

- **Materials:**

1. Raw Data from Data Controller in Natural Gas Company.

6. Approval

- **Instructor/Advisor:** Dr / Mohamed El-Najjar
- **Signature:**